Tutorial No : 6

Tutorial Title: Implement Stack using Linked List
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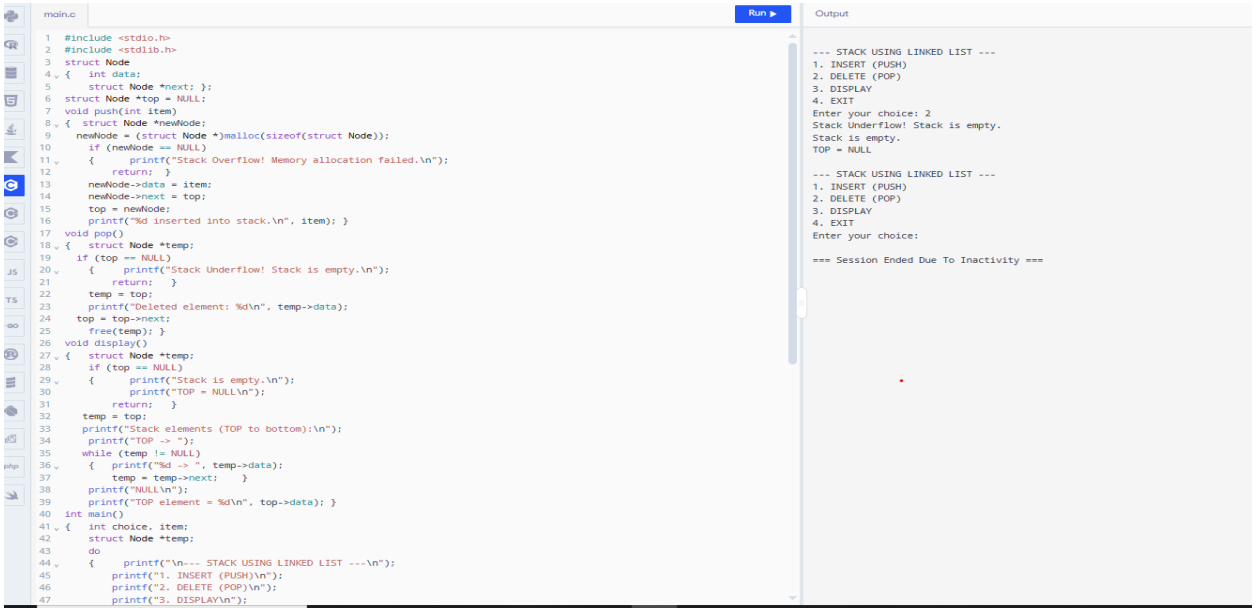
Program:

```
#include <stdio.h>
#include <stdlib.h>
struct Node
{   int data;
   struct Node *next; };
struct Node *top = NULL;
void push(int item)
{  struct Node *newNode;
  newNode = (struct Node *)malloc(sizeof(struct Node));
   if (newNode == NULL)
   {      printf("Stack Overflow! Memory allocation failed.\n");
      return;  }
   newNode->data = item;
   newNode->next = top;
   top = newNode;
   printf("%d inserted into stack.\n", item); }
void pop()
{   struct Node *temp;
  if (top == NULL)
   {     printf("Stack Underflow! Stack is empty.\n");
      return;   }
   temp = top;
   printf("Deleted element: %d\n", temp->data);
  top = top->next;
   free(temp); }
void display()
{   struct Node *temp;
   if (top == NULL)
   {      printf("Stack is empty.\n");
         printf("TOP = NULL\n");
      return;   }
```

```
  temp = top;
  printf("Stack elements (TOP to bottom):\n");
   printf("TOP -> ");
  while (temp != NULL)
   {   printf("%d -> ", temp->data);
      temp = temp->next;    }
   printf("NULL\n");
   printf("TOP element = %d\n", top->data); }
int main()
{   int choice, item;
   struct Node *temp;
   do
   {     printf("\n--- STACK USING LINKED LIST ---\n");
      printf("1. INSERT (PUSH)\n");
      printf("2. DELETE (POP)\n");
      printf("3. DISPLAY\n");
      printf("4. EXIT\n");
      printf("Enter your choice: ");
     if (scanf("%d", &choice) != 1)
          break;
     switch (choice)
      {       case 1:
         printf("Enter element to insert: ");
        if (scanf("%d", &item) != 1)
             {    choice = 4;
                break;    }
            push(item);
             display();
             break;
            case 2:
             pop();
             display();
             break;
             case 3:
             display();
             break;
              case 4:
             printf("Exiting program.\n");
             break;
               default:
```

```
            printf("Invalid choice!\n");       }
  } while (choice != 4);
  while (top != NULL)
  {     temp = top;
      top = top->next;
      free(temp);   }
 return 0; }
```

Output:



main.c

Run ▶

```c
    printf("TOP -> ");
    while (temp != NULL)
    {   printf("%d -> ", temp->data);
        temp = temp->next;    }
    printf("NULL\n");
    printf("TOP element = %d\n", top->data); }
int main()
{   int choice, item;
    struct Node *temp;
    do
    {     printf("\n--- STACK USING LINKED LIST ---\n");
        printf("1. INSERT (PUSH)\n");
        printf("2. DELETE (POP)\n");
        printf("3. DISPLAY\n");
        printf("4. EXIT\n");
        printf("Enter your choice: ");
      if (scanf("%d", &choice) != 1)
            break;
      switch (choice)
        {       case 1:
          printf("Enter element to insert: ");
         if (scanf("%d", &item) != 1)
                {    choice = 4;
                    break;    }
              push(item);
               display();
               break;
              case 2:
               pop();
               display();
               break;
               case 3:
               display();
               break;
                case 4:
               printf("Exiting program.\n");
               break;
                 default:
               printf("Invalid choice!\n");       }
    } while (choice != 4);
    while (top != NULL)
    {     temp = top;
        top = top->next;
        free(temp);   }
   return 0; }
```

Output

```
--- STACK USING LINKED LIST ---
1. INSERT (PUSH)
2. DELETE (POP)
3. DISPLAY
4. EXIT
Enter your choice: 2
Stack Underflow! Stack is empty.
Stack is empty.
TOP = NULL

--- STACK USING LINKED LIST ---
1. INSERT (PUSH)
2. DELETE (POP)
3. DISPLAY
4. EXIT
Enter your choice:

=== Session Ended Due To Inactivity ===
```